

| Game                | Human           | IW(1)             | IW(1)       |                  | Rollout IW(1) |                | RA Rollout IW(1) |                  | RAS Rollout IW(1) |                   |
|---------------------|-----------------|-------------------|-------------|------------------|---------------|----------------|------------------|------------------|-------------------|-------------------|
|                     |                 | RAM               | budget .5s  | budget 32s       | budget .5s    | budget 32s     | budget .5s       | budget 32s       | budget .5s        | budget 32s        |
| alien               | 6,875.0         | <b>25,634.0</b>   | 1,316.0     | 14,010.0         | 4,238.0       | 6,896.0        | 7,170.0          | 13,454.0         | 8,550.0           | 19,354.0          |
| amidar              | 1,676.0         | 1,377.0           | 48.0        | 1,043.2          | 659.8         | 1,698.6        | 1,049.2          | <b>1,794.0</b>   | 1,161.0           | 1,609.0           |
| assault             | <b>1,496.0</b>  | 953.0             | 268.8       | 336.0            | 285.6         | 319.2          | 336.0            | 327.6            | 264.6             | 281.4             |
| asterix             | 8,503.0         | 153,400.0         | 1,350.0     | <b>262,500.0</b> | 45,780.0      | 66,100.0       | 46,100.0         | 67,100.0         | 48,700.0          | 87,600.0          |
| asteroids           | 13,157.0        | <b>51,338.0</b>   | 840.0       | 7,630.0          | 4,344.0       | 7,258.0        | 4,698.0          | 6,836.0          | 4,486.0           | 7,344.0           |
| atlantis            | 29,028.0        | <b>159,420.0</b>  | 33,160.0    | 82,060.0         | 64,200.0      | 151,120.0      | 122,220.0        | 134,660.0        | 113,460.0         | 134,660.0         |
| bank heist          | 734.4           | 717.0             | 24.0        | 739.0            | 272.0         | 865.0          | 242.0            | 1,323.4          | 268.0             | <b>2,179.0</b>    |
| battle zone         | 37,800.0        | 11,600.0          | 6,800.0     | 14,800.0         | 39,600.0      | 414,000.0      | 74,600.0         | 455,800.0        | 56,200.0          | <b>509,400.0</b>  |
| beam rider          | 5,775.0         | <b>9,108.0</b>    | 715.2       | 1,530.4          | 2,188.0       | 2,464.8        | 2,552.8          | 5,367.2          | 3,729.2           | 4,921.2           |
| berzerk             | n/a             | <b>2,096.0</b>    | 280.0       | 1,318.0          | 644.0         | 862.0          | 1,208.0          | 1,454.0          | 966.0             | 1,640.0           |
| bowling             | <b>154.8</b>    | 69.0              | 30.6        | 49.2             | 47.6          | 45.8           | 44.2             | 49.0             | 51.6              | 48.0              |
| boxing              | 4.3             | <b>100.0</b>      | 99.4        | 79.0             | 75.4          | 79.4           | 99.2             | <b>100.0</b>     | 78.6              | 80.2              |
| breakout            | 31.8            | <b>384.0</b>      | 1.6         | 56.0             | 82.4          | 36.0           | 86.2             | 336.4            | 79.8              | 370.0             |
| centipede           | 11,963.0        | 99,207.0          | 88,890.0    | <b>143,275.4</b> | 36,980.2      | 65,162.6       | 56,328.0         | 92,353.0         | 46,661.4          | 84,226.0          |
| chopper command     | 9,882.0         | 10,980.0          | 1,760.0     | 1,800.0          | 2,920.0       | 5,800.0        | 9,820.0          | 11,240.0         | 8,900.0           | <b>33,220.0</b>   |
| crazy climber       | 35,411.0        | 36,160.0          | 16,780.0    | <b>44,340.0</b>  | 39,220.0      | 43,960.0       | 40,440.0         | 38,460.0         | 38,120.0          | 42,720.0          |
| defender            | n/a             | n/a               | 362,010.0   | <b>430,010.0</b> | 373,010.0     | 256,010.0      | 371,010.0        | 374,010.0        | 298,010.0         | 387,010.0         |
| demon attack        | 3,401.0         | 20,116.0          | 106.0       | <b>23,619.0</b>  | 2,780.0       | 9,996.0        | 6,958.0          | 10,753.0         | 5,201.0           | 9,898.0           |
| double dunk         | -15.5           | -14.0             | -22.0       | -22.4            | 3.6           | <b>20.0</b>    | 3.2              | 19.6             | -4.0              | 16.0              |
| elevator action     | n/a             | <b>13,480.0</b>   | 1,080.0     | 0.0              | 0.0           | 0.0            | 0.0              | 0.0              | 0.0               | 1,300.0           |
| enduro              | 309.6           | <b>500.0</b>      | 2.6         | 229.2            | 169.4         | 359.4          | 145.8            | 381.0            | 137.4             | 330.8             |
| fishing derby       | 5.5             | <b>30.0</b>       | -83.8       | -39.0            | -68.0         | -16.2          | -77.0            | -50.2            | -61.8             | -53.0             |
| freeway             | 29.6            | <b>31.0</b>       | 0.6         | 25.0             | 2.8           | 12.6           | 2.0              | 11.2             | 3.6               | 10.0              |
| frostbite           | 4,335.0         | 902.0             | 106.0       | 182.0            | 220.0         | 5,484.0        | 146.0            | <b>6,398.0</b>   | 1,494.0           | 5,970.0           |
| gopher              | 2,321.0         | 18,256.0          | 1,036.0     | <b>18,472.0</b>  | 7,216.0       | 13,176.0       | 8,388.0          | 13,144.0         | 7,256.0           | 11,840.0          |
| gravitar            | 2,672.0         | 3,920.0           | 380.0       | 1,630.0          | 1,630.0       | 3,700.0        | 1,660.0          | 5,130.0          | 2,410.0           | <b>5,540.0</b>    |
| hero                | 25,673.0        | 12,985.0          | 2,034.0     | 7,432.0          | 13,709.0      | 28,260.0       | 11,377.0         | 24,072.0         | 11,480.0          | <b>29,708.0</b>   |
| ice hockey          | 0.9             | <b>55.0</b>       | -13.6       | -7.0             | -6.0          | 6.6            | -12.4            | -2.6             | 5.2               | 18.2              |
| james bond          | 406.7           | <b>23,070.0</b>   | 40.0        | 180.0            | 450.0         | 22,250.0       | 10,760.0         | 12,656.0         | 5,340.0           | 12,345.0          |
| kaboom              | n/a             | n/a               | 127.0       | 178.2            | 87.4          | 159.6          | 23.2             | 151.6            | 39.8              | <b>255.6</b>      |
| kangaroo            | 3,035.0         | <b>8,760.0</b>    | 160.0       | 3,820.0          | 1,080.0       | 5,780.0        | 1,880.0          | 4,600.0          | 1,800.0           | 5,280.0           |
| krull               | 2,395.0         | <b>6,030.0</b>    | 3,206.8     | 5,611.8          | 1,892.8       | 1,151.2        | 2,091.8          | 2,219.8          | 1,645.2           | 2,837.0           |
| kung fu master      | 22,736.0        | <b>63,780.0</b>   | 440.0       | 8,980.0          | 2,080.0       | 14,920.0       | 2,620.0          | 26,540.0         | 2,980.0           | 24,300.0          |
| montezuma's revenge | <b>4,367.0</b>  | 0.0               | 0.0         | 0.0              | 0.0           | 0.0            | 100.0            | 1,620.0          | 100.0             | 1,080.0           |
| ms pacman           | 15,693.0        | 21,695.0          | 2,578.0     | 20,622.8         | 9,178.4       | 19,667.0       | 15,115.0         | <b>23,033.0</b>  | 13,746.8          | 21,833.0          |
| name this game      | 4,076.0         | 9,354.0           | 7,070.0     | <b>13,478.0</b>  | 6,226.0       | 5,980.0        | 6,558.0          | 6,870.0          | 6,128.0           | 6,820.0           |
| phoenix             | n/a             | n/a               | 1,266.0     | 5,550.0          | 5,750.0       | <b>7,636.0</b> | 6,790.0          | 7,460.0          | 5,386.0           | 7,570.0           |
| pitfall             | n/a             | n/a               | <b>-8.6</b> | -92.2            | -81.4         | -130.8         | -302.8           | -723.4           | -814.6            | -692.4            |
| pong                | 9.3             | <b>21.0</b>       | -20.8       | 0.8              | -7.4          | 17.6           | -4.2             | 15.2             | -1.4              | 18.2              |
| private eye         | <b>69,571.0</b> | -99.0             | 2,690.8     | -526.4           | -322.0        | 3,157.2        | -480.0           | -300.0           | 2,160.0           | -340.0            |
| qbert               | 13,455.0        | 3,705.0           | 515.0       | 16,505.0         | 3,375.0       | 8,390.0        | 15,970.0         | 26,875.0         | 14,160.0          | <b>40,350.0</b>   |
| riveraid            | 13,513.0        | 5,694.0           | 664.0       | 7,042.0          | 6,088.0       | 8,156.0        | 6,288.0          | 14,700.0         | 7,138.0           | <b>15,302.0</b>   |
| road runner         | 7,845.0         | <b>94,940.0</b>   | 200.0       | 0.0              | 2,360.0       | 37,080.0       | 31,140.0         | 44,020.0         | 25,780.0          | 62,960.0          |
| robotank            | 11.9            | <b>68.0</b>       | 3.2         | 32.8             | 31.0          | 52.6           | 31.2             | 47.4             | 30.0              | 51.8              |
| sequest             | <b>20,182.0</b> | 14,272.0          | 168.0       | 356.0            | 980.0         | 10,932.0       | 2,312.0          | 13,472.0         | 1,236.0           | 9,846.0           |
| skiing              | n/a             | n/a               | -16,511.0   | -15,962.0        | -15,738.8     | -16,477.0      | -16,006.8        | <b>-15,244.2</b> | -15,806.6         | -15,473.6         |
| solaris             | n/a             | n/a               | 1,356.0     | <b>2,300.0</b>   | 700.0         | 1,040.0        | 1,704.0          | 692.0            | 1,620.0           | 1,728.0           |
| space invaders      | 1,652.0         | 2,877.0           | 280.0       | 1,963.0          | 2,628.0       | 1,980.0        | 1,149.0          | 2,533.0          | 1,812.0           | <b>4,362.0</b>    |
| star gunner         | 10,250.0        | 1,540.0           | 840.0       | 1,340.0          | 13,360.0      | 15,640.0       | 14,900.0         | 16,460.0         | 15,960.0          | <b>17,160.0</b>   |
| tennis              | -8.9            | <b>24.0</b>       | -23.4       | -22.2            | -18.6         | -2.2           | -5.4             | 14.8             | 3.2               | -3.4              |
| time pilot          | 5,925.0         | <b>35,000.0</b>   | 2,360.0     | 5,740.0          | 7,640.0       | 8,140.0        | 3,540.0          | 5,480.0          | 8,540.0           | 5,480.0           |
| tutankham           | 167.7           | 172.0             | 71.2        | 172.4            | 128.4         | 184.0          | 135.6            | <b>191.6</b>     | 147.4             | 191.2             |
| up n down           | 9,082.0         | <b>110,036.0</b>  | 928.0       | 62,378.0         | 36,236.0      | 44,306.0       | 34,668.0         | 39,964.0         | 36,936.0          | 41,056.0          |
| venture             | 1,188.0         | <b>1,200.0</b>    | 0.0         | 240.0            | 0.0           | 80.0           | 60.0             | 500.0            | 80.0              | 120.0             |
| video pinball       | 17,298.0        | 388,712.0         | 28,706.4    | <b>441,094.2</b> | 203,765.4     | 382,294.8      | 216,468.6        | 378,815.4        | 188,604.4         | 375,073.0         |
| wizard of wor       | 4,757.0         | <b>121,060.0</b>  | 5,660.0     | 115,980.0        | 37,220.0      | 73,820.0       | 43,860.0         | 84,660.0         | 40,780.0          | 75,380.0          |
| yars revenge        | n/a             | n/a               | 6,352.6     | 10,808.2         | 5,225.4       | 9,866.4        | 7,848.8          | <b>23,261.8</b>  | 3,647.8           | 10,523.6          |
| zaxxon              | 9,173.0         | 29,240.0          | 0.0         | 15,080.0         | 9,280.0       | 22,880.0       | 15,500.0         | 34,180.0         | 18,700.0          | <b>38,700.0</b>   |
| # ≥ Human           | n/a             | n/a               | 7 (14.2%)   | 22 (44.8%)       | 19 (38.7%)    | 34 (69.3%)     | 22 (44.8%)       | 35 (71.4%)       | 25 (51.0%)        | <b>37 (75.5%)</b> |
| # ≥ 75% Human       | n/a             | n/a               | 7 (14.2%)   | 24 (48.9%)       | 22 (44.8%)    | 34 (69.3%)     | 26 (53.0%)       | 39 (79.5%)       | 29 (59.1%)        | <b>40 (81.6%)</b> |
| # best in game      | 5 (8.6%)        | <b>24 (41.3%)</b> | 1 (1.7%)    | 9 (15.5%)        | 0 (0.0%)      | 2 (3.4%)       | 0 (0.0%)         | 7 (12.0%)        | 0 (0.0%)          | 11 (18.9%)        |

Table 2: Scores over 58 games for Human, IW(1) over RAM, and IW(1) and three versions of Rollout IW(1) with time budgets of 0.5 and 32 seconds. Entries marked as n/a are not available, those highlighted in bold are best scores across rows, and those highlighted in red are scores as good as human playing (except for IW(1) over RAM). The 9 games for which there are no reports for the human player are not considered when comparing the different algorithms with the human player.

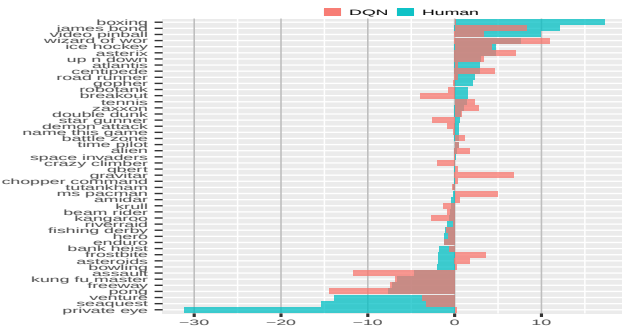


Figure 3: Comparison of RAS Rollout IW(1) with time budget of 0.5 seconds with the human player and DQN using relative scores over 48 games. Blue bars depict RAS Rollout IW(1) scores ( $r$ ) vs. human scores ( $h$ ), while red bars, RAS Rollout IW(1) scores vs. DQN scores ( $d$ ). The bar has positive length  $(r - h)/h$  when  $r > h$ , and negative length  $(h - r)/r$  when  $r < h$ . Bars have zero length when  $r = h$ . Same for  $d$  in place of  $h$ .

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